Northern Devon Healthcare NHS Trust

Document Control

Title **Management of Fever and Neutropenia (Neutropenic Sepsis) in Paediatric Oncology Patients Presenting to NDDH, Guidelines**	
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1.1	Jan 2014	Final	Addition of appendix 3 – summary of guidelines
1.2	Feb 2013	Final	Approved by DTC with minor amendments.
2.0	Mar 14	Final	Published on Bob
2.1	Oct 15	Revision	New template, adjustment to dosing interval as per RD+E guidance (derived from BNFc), title amended to include "neutropenic sepsis"
2.2	Sept 18	Update	Sections 1.1 and 1.2 added Guidance on blood transfusions altered to align with the NDDH hospital transfusion policy Section 7.3 on investigation results added Useful numbers and list of Bristol Children's Hospital Guidelines updated. Ratified by DTC 20th Sept 2018.
3.0	Sept 18	FINAL	Ratified by DTC 20th Sept 2018.

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Superseded Documents Management of Fever and Neutropenia in Paediatric Oncology Patients presenting to NDDH Guidelines v2.1	

Issue Date	Review Date	Review Cycle
Sept 18	Sept 2021	Three years

Consulted with the following stakeholders:

Antibiotic Working Group
Drug & Therapeutics Group
Paediatricians
Haematologists

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CONTENTS

Document Control **1**
1. Purpose **4**
2. Definitions **5**
Febrile Neutropenia (Neutropenic Sepsis) 5
3. Contact Numbers **5**
4. Investigation of Fever in Paediatric Oncology Patients **6**
4.1 Initial investigations of the febrile neutropenic patient 6
4.2 Baseline investigations (see section 5.3 for suspected line sepsis) 6
5. Empirical Treatment of Febrile Neutropenia **7**
6. Line Sepsis **9**
7. Management of Febrile Neutropenia **10**
8. Monitoring Compliance with and the Effectiveness of the Policy **12**
9. Equality Impact Assessment **13**
10. References **14**
11. Associated Documentation **14**
12. Appendix 1 – Formula for Calculating the Blood volume to transfuse **15**
13. Appendix 2 – Clinical Assessment Audit Form **16**
14. Appendix 3 – Guideline Summary **17**

1. Purpose

1.1. Local patients with a paediatric oncology condition are under shared care, their principal treatment centre is Bristol Children's Hospital and their paediatric oncology shared care unit is the Royal Devon and Exeter Hospital (R,D and E). Patients have open access to Caroline Thorpe Ward and are asked to attend NDDH for assessment and treatment if neutropenic sepsis is a possibility to avoid the time delay there would be if they went straight to the R,D and E. Patients are admitted under the care of the attending Paediatric Consultant at NDDH, decisions regarding their management are made with their Paediatric Oncology Team.

1.2. Patients with a potential new diagnosis of a Paediatric Oncology condition should be assessed on Caroline Thorpe Ward as soon as the referral is made and then discussed with the Paediatric Oncology Consultant at Bristol Children's Hospital. The Bristol Consultant will advise regarding any treatment to commence and investigations to complete prior to transfer

1.3. The purpose of this document is to detail the process for management of febrile neutropenia and neutropenic sepsis in children (aged less than or equal to 16 years), who are neutropenic or immune-compromised as a result of chemotherapy in the acute hospital setting. All children undergoing chemotherapy/radiotherapy are immuno-compromised to some degree, whatever their blood count.

1.4. The guideline applies to all prescribing teams in the acute hospital and must be adhered to. Other staff (e.g. Nursing staff, Pharmacists) may need to familiarise themselves with some aspects of the guideline. Non-compliance with this guideline may be for valid clinical reasons only. The reason for non-compliance must be documented in the patient's notes.

1.5. Implementation of this best practice guideline will ensure that:

Antibiotic prescribing follows the principles of good antibiotic stewardship
Investigation of infection is appropriate and timely

2. Definitions

Febrile Neutropenia (Neutropenic Sepsis)

Neutrophils less than 0.5 x 109/L and fever greater than 38°C (assessed by any means)

OR

Clinical suspicion of sepsis in the absence of fever, e.g. unexplained abdominal pain or generally unwell.

Given the large geographical area covered by our service, we instruct some patients to report lower fevers (e.g. greater than 37.5°C). Generally we instruct patients to come directly to the ward and if their temperature remains below 38°C and they are well, antibiotics can be withheld (if in doubt, discuss with the duty Oncology Consultant).

3. Contact Numbers

North Devon District Hospital (NDDH)

Microbiologist:	Bleep 193. Via switchboard out of hours
Antibiotic Pharmacist:	Bleep 029
Jenny Birch (CLIC Nurse)	0776 7308469
Consultant Paediatrician	Bleep 340. Via Switchboard out of hours

Royal Devon and Exeter Hospital (RD&E)

Dr Corinne Hayes or Dr Simon Parke	01392 402684/ 402685
Or via Bramble Ward	01392 402679/ 402678
Oncology team E-mail:	rde-tr.PaediatricOncology@nhs.net

Out of hours first point of contact (RD&E)

On call Paediatric Registrar, via Bramble Yellow Ward, also liaise with Paediatric Oncology Nurse on shift at Bramble Ward, RD&E

Bristol Children's Hospital 0117 9230000

Contact Oncology Day Beds, or on call Oncology Registrar/ Consultant

4. Investigation of Fever in Paediatric Oncology Patients

4.1 Initial investigations of the febrile neutropenic patient

Full physical examination including the following:

- Examination of central venous access device for exit-site or tunnel infection
- Examination of mouth for mucositis
- ENT examination
- Assessment of nappy area / perineum

4.2 Baseline investigations (see section 5.3 for suspected line sepsis)

- FBC & differential (purple bottle)
- Routine biochemistry & CRP (brown bottle)
- Lactate (use gas machine on SCBU or yellow bottle and inform biochemistry laboratory)
- Blood cultures from each lumen of central venous line, or peripherally (only if no central line)
- CXR (if clinical suspicion of respiratory focus)
- Urine dipstick and culture for all ages (do NOT delay starting antibiotics for urine sample)
- Swabs from sites of clinical infection only (always moisten swabs before taking sample)
    - **For fever with acute diarrhoea and / or vomiting, add:**
        - Stool for viral studies (blue form),
        - MC&S, C. difficile toxin, ova, cysts and parasites (blue form)
    - **For fever with respiratory symptoms add viral throat swab and / or an NPA-request "respiratory panel"**
        - These swabs only need to be done if the patient is NOT being transferred to the RD&E (they are processed more quickly if sent from there).

NB. Ensure forms have "neutropenic" written in clinical details

5. Empirical Treatment of Febrile Neutropenia

If the patient is "ill", treatment must be given immediately by the admitting doctor even in the absence of a fever or neutropenia

DELAY IN STARTING ANTIBIOTICS MAY PROVE FATAL.
ANTIBIOTICS MUST BE STARTED AS SOON AS POSSIBLE
AIM FOR 100% PATIENTS TO BE GIVEN ANTIBIOTICS WITHIN 30 MINUTES.

Usual treatment duration: 5-7 days

NOTE: Some patients may have a hand-held "patient-specific protocol" for antibiotic management and / or a plan for use of their portacath – please check with the parents. Information is also in the "weekly Oncology MDT minutes" e-mailed to all Paediatric Consultants and saved in the Paediatric Drive – Oncology MDT minutes (e.g. avoid Piperacillin-tazobactam (Tazocin®) during treatment with high dose methotrexate).

Check recent culture results e.g. MRSA, ESBLs before prescribing empirically. Discuss with Paediatric Oncology Consultant if in doubt.

5.1. Empirical Treatment (first line)

- Commence (as soon as possible, within 30 minutes)
- **Piperacillin-tazobactam (Tazocin®) 90mg/kg/dose 6 hourly IV** (maximum single dose 4.5g)
- **Add vancomycin 15mg/kg/dose 8 hourly IV** (maximum daily dose 2g) if clinical evidence or suspicion of line sepsis or tunnel infection (see below)
- Check levels before 3rd dose (aim for trough 10-15mg/L). Infuse over 90 minutes.
- **Consider adding a single dose of gentamicin IV (7mg/kg based on ideal body weight)** if concern regarding severe sepsis from clinical assessment (e.g. persistent tachycardia, raised lactate, low albumin etc)

5.2. Empirical treatment (mild – moderate penicillin allergy)

- Commence (as soon as possible, within 30 minutes)
- **Meropenem 20mg/kg/dose 8 hourly IV** (maximum single dose 2g)

- **Add vancomycin 15mg/kg/dose 8 hourly IV** (maximum daily dose 2g) if clinical evidence or suspicion of line sepsis or tunnel infection (see below)

5.3. Check levels before 3rd dose (aim for trough 10-15mg/L). Infuse over 90 minutes.

- **Consider adding a single dose of gentamicin IV (7mg/kg based on ideal body weight)** if concern regarding severe sepsis from clinical assessment (e.g. persistent tachycardia, raised lactate, low albumin etc)

5.4. Empirical treatment (severe penicillin allergy)
Commence (as soon as possible, within 30 minutes)

- **Aztreonam IV (see BNFc for dose) and vancomycin IV**

- **Consider adding a single dose of gentamicin IV (7mg/kg based on ideal body weight)** if concern regarding severe sepsis from clinical assessment (e.g. persistent tachycardia, raised lactate, low albumin etc)

5.5. **Review all antibiotics at 48 hours, or sooner if culture results are available**

6. Line Sepsis

6.1. Take bloods (as per section 4) and cultures from all lumens of central line.

6.2. Swab exit site.

6.3. If line sepsis is suspected e.g. fever and rigors after flushing line, take blood cultures and start empirical antibiotics (as per section 5), irrespective of neutrophil count and await cultures.

6.4. If the patient is systemically well but has a red/ mucky exit site:

- Use a moistened swab to sample the exit site
- Take blood cultures
- Give oral flucloxacillin 1st line, or oral clindamycin if penicillin allergy for 5-7 days (see BNFc for up to date dose recommendations)
- Observe the patient for 45 minutes post line flush before discharge

7. Management of Febrile Neutropenia

7.1. Ongoing management

- Daily clinical examination, CRP and reassessment.
- In general, all cases should be discussed with Microbiology and you must ensure any positive results are discussed with them

7.2. Stopping antibiotics

- The decision to stop antibiotics should always be discussed with a consultant before discontinuing a prescription.
- All known oncology patients who are seen should be discussed with both the NDDH Paediatric Consultant On-Call and either Dr Hayes or Dr Parke at the RD&E (contactable via Bramble Ward), or if they are not available then contact the Paediatric Oncology team at Bristol Children's Hospital (any new patients presenting with an oncological diagnosis should be discussed with the Bristol team).
- Useful contact numbers are listed in section 3
- When the patient has been stabilised, appropriate transfer (usually to the RD&E) needs to be arranged. These patients are at risk of life threatening sepsis therefore transfer needs to be by ambulance with, at least, a paramedic crew as an escort.

7.3. Other considerations

R, D and E Paediatric Oncology team MDT minutes

- Useful information about our shared Oncology patients is in the minutes of the weekly RD&E MDT (multi- disciplinary team meeting) minutes which are saved on the Paediatric Drive, Oncology MDT file and e-mailed out to all consultants and to the Caroline Thorpe Ward Sister and the Paediatric Community Nursing Lead .

Blood and Platelet Transfusions

- Depending on the results of blood tests, blood and platelet transfusions may be necessary. Please give our blood transfusion laboratory as much notice as possible. The specification of blood products needed for some patients is held in our transfusion laboratory. There may be additional information regarding patient specific transfusion requirements in the minutes of the R,D and E Oncology MDT meetings, if there are any queries regarding the type of blood to transfuse liaise with the R,D and E Oncology team. See Appendix 1: Formula for calculating the blood volume to transfuse.

Investigation results

- On transfer to the R,D and E (or to Bristol Children's Hospital / discharge home) patients should be sent with a discharge summary listing all treatments they have had and all investigations that have been taken including all results that are back. Results that subsequently come back should be scanned and e-mailed to the R,D and E Oncology team using their generic e-mail. Any positive culture results or abnormal haematology / biochemistry not included in the patients discharge summary should be relayed to the Oncology Nurse on shift at the R,D and E by phone as soon as they are available and then also e-mailed as above.

8. Monitoring Compliance with and the Effectiveness of the Policy

8.1. Standards/ Key Performance Indicators

Adherence to guidelines may be audited by the microbiology and paediatrics departments. Critical incident reports relating to infection will be collated by the antibiotic pharmacist. Results will be reported on an annual basis to the Antibiotic Working Group.

Suggested audit standards include:

- Clinical Assessment audit (based on NICE clinical guideline 151) – see appendix 2
- All patients with febrile neutropenia should be administered an appropriate antimicrobial agent within 1 hour of presentation.

9. Equality Impact Assessment

The author must include the Equality Impact Assessment Table and identify whether the policy has a positive or negative impact on any of the groups listed. The Author must make comment on how the policy makes this impact.

Table 1: Equality impact Assessment

Group	Positive Impact	Negative Impact	No Impact	Comment
Age	×			Specialist guidance for quality improvement in paediatric prescribing
Disability			×	
Gender			×	
Gender Reassignment			×	
Human Rights (rights to privacy, dignity, liberty and non-degrading treatment)			×	
Marriage and civil partnership			×	
Pregnancy		×		Individual patients must be assessed for risk, some medicinal products recommended may harm the unborn foetus. It may be appropriate to treat outside of guidance using local microbiologist advice.
Maternity and Breastfeeding		×		Individual patients must be assessed for risk, some medicinal products recommended may harm the infant. It may be appropriate to treat outside of guidance using local microbiologist advice.
Race (ethnic origin)			×	
Religion (or belief)			×	

Sexual Orientation			×	

10. References

Royal Devon and Exeter protocol for management of neutropenic sepsis in paediatric patients (2015)

Neutropenic sepsis: prevention and management of neutropenic sepsis in cancer patients. NICE clinical guideline 151. Issued: September 2012, reviewed 2014.

Paediatric Formulary Committee. BNF for Children (online) London: BMJ Group, Pharmaceutical Press, and RCPCH Publications https://www.new.medicinescomplete.com/

11. Associated Documentation

Bristol Children's Hospital Protocols on the Following Oncological emergencies may be helpful:

- Acutely Unwell Haem/Onc patient Management
- Brain and Spinal Cord Tumour Management
- Encephalopathy Management in Haem/Onc patients
- Haematology and Oncology New Patient Checklist
- Leukaemia: management of initial high WCC
- Spinal Cord Compression Management in Haem/Onc patients
- SVC Obstruction Management in Haem/Onc patients
- Tumour Lysis Syndrome

12. Appendix 1 – Formula for Calculating the Blood volume to transfuse

The volume of packed red cells to be transfused should be calculated according to the formula below and always prescribed in mls not units:

Weight of the patient (kg) X Rise of Hb (g/L) ÷ 10 X 4 = Volume to be Transfused (mls)
Establish with a senior doctor the aimed for final Hb concentration and current Hb.

e.g. Calculation for rise of Hb needed if current Hb 70g/L and aimed for Hb 120g/L:

120 - 70 = 50g/L is the aimed for rise.

15kg (patient weight) x 50g/L (rise of Hb) x 4 = 300mls to be transfused

Do not give more than 20ml/kg per transfusion unless actively bleeding.

Red cells should be transfused within 4 hours following removal from storage.

Discuss with the lab the number of mls available in each unit (which can vary). Try to limit the number of units used if the transfusion volume is close (approx. within 20mls) to the unit size available. Paediatric sized administration sets are available for transfusing small volumes.

Patient and product checks must be carried out by two registered staff who must complete their own independent checks prior to setting up a transfusion.

Taken from 'Transfusions for Children' NDDH guideline

13. Appendix 2 – Clinical Assessment Audit Form

Data collection form for Neutropenic sepsis: assessment clinical audit

Audit ID:	Sex:	Age:

The audit ID should be an anonymous code. Patient identifiable information should never be recorded.

No	Question	Yes	No	Exception* /NA/Notes
Assessment				
1	Was an initial clinical assessment carried out?	☐	☐	
	If yes, did it include the following?			
	history	☐	☐	
	examination	☐	☐	
	full blood count	☐	☐	
	kidney function test	☐	☐	
	liver function test	☐	☐	
	– including albumin	☐	☐	
	C-reactive protein	☐	☐	
	Lactate	☐	☐	
	blood culture.	☐	☐	
2	Were appropriate antibiotics given within 1 hour of hospital assessment?	☐	☐	

*Circle exception codes as appropriate.

Exception codes

A – peripheral blood culture is not clinically feasible.
B – patient doesn't have a central venous access device.
C – patient is aged 5 years or over

14. Appendix 3 – Guideline Summary

Summary of Guidelines for Management of Fever and Neutropenia in Paediatric Oncology Patients presenting to NDDH

All children undergoing chemotherapy/radiotherapy are immuno-compromised to some degree, whatever their blood count.

If the patient is "ill", treatment must be given immediately by the admitting doctor even in the absence of a fever or neutropenia

DELAY IN STARTING ANTIBIOTICS MAY PROVE FATAL.
ANTIBIOTICS MUST BE STARTED AS SOON AS POSSIBLE
AIM FOR 100% PATIENTS TO BE GIVEN ANTIBIOTICS WITHIN 30 MINUTES.

Definition of Febrile Neutropenia

- Neutrophils <0.5 x 109/L and
- Fever >**38°C** (assessed by any means)
- OR clinical suspicion of sepsis in the absence of fever, e.g. unexplained abdominal pain or generally unwell.

Given the large geographical area covered by our service we instruct some patients to report lower fevers, e.g. >37.5°C. Generally we instruct patients to come directly to the ward and if their temperature remains <38°C and they are well, antibiotics can be withheld. (If in doubt discuss with the duty Oncology Consultant)

Initial Investigations of the Febrile Neutropenic Patient

- Full physical examination including the following:
- Examination of central venous access device for exit-site or tunnel infection
- Examination of mouth for mucositis
- ENT examination
- Assessment of nappy area / perineum

Baseline investigations: (see 4 for suspected line sepsis)

- FBC & differential (purple bottle)
- Routine biochemistry & CRP (brown bottle)
- Lactate (use gas machine on SCBU or yellow bottle and inform biochemistry laboratory)
- Blood cultures from each lumen of central venous line, or peripherally (only if no central line)
- CXR (if clinical suspicion of respiratory focus)
- Urine dipstick and culture for all ages (do NOT delay starting antibiotics for urine sample)

- Swabs from sites of clinical infection only (always moisten swabs before taking sample)

For fever with acute diarrhoea and/or vomiting add

- Stool for viral studies (blue form),
- M,C&S, C. diff toxin, ova, cysts and parasites (blue form)

For fever with respiratory symptoms add viral throat swab and / or an NPA-request 'respiratory panel'

- These swabs only need to be done if the patient is NOT being transferred to the R, D and E (they are processed more quickly if sent from there)

(Ensure forms have "neutropenic" written on clinical details)

Empirical Treatment of Febrile Neutropenia

Commence (@ T=0 hours):

NOTE: Some patients may have a hand-held "patient-specific protocol" for antibiotic management – please check with the parents. If so, a copy should be in the alert box in the notes as well as the weekly oncology MDT minutes saved on the Paediatric Drive (e.g. avoid Piperacillin/tazobactam (Tazocin®) during treatment with high dose methotrexate).

Check recent culture results e.g. MRSA, ESBLs before prescribing empirically. Discuss with Paediatric Haematology Consultant if in doubt.

Empirical Treatment:

- **First line: Piperacillin-tazobactam** 90mg/kg/dose 6 hourly (maximum dose 4.5g)

OR

- **Mild/moderate penicillin allergy**: Meropenem 20mg/kg/dose 8 hourly (maximum single dose 2g)

OR

- **Penicillin anaphylaxis: Aztreonam** (see BNFc for doses) **AND vancomycin** 15mg/kg/dose 8 hourly (maximum daily dose 2g) – check levels before 3rd dose (aim for trough 10-20mg/L). Infuse over 90 minutes.
- **Line infection: Add Vancomycin** 15mg/kg/dose 8 hourly if not already prescribed (maximum daily dose 2 g) if clinical evidence or suspicion of line sepsis or tunnel infection (see below) – check levels before 3rd dose (aim for trough 10-15mg/L). Infuse over 90 minutes
- **Severe sepsis: Consider adding a single dose of gentamicin** (7mg/kg based on ideal body weight) if concern regarding severe sepsis from clinical assessment e.g. persistent tachycardia, raised lactate, low albumin etc

Line Sepsis/ Tunnel Infection

- Take bloods (as per previous page) and blood cultures from all lumens of central line.
- Swab exit site.
- If line sepsis is suspected e.g. fever and rigors after flushing line, take blood cultures and start empirical antibiotics irrespective of neutrophil count and await cultures. (Piperacillin-tazobactam/meropenem/aztreonam and Vancomycin are indicated for suspected line sepsis).
- If the patient is systemically well but has a red/ mucky exit site, use a moistened swab to sample the exit site, take blood cultures and give oral flucloxacillin (or clindamycin in penicillin-allergic patients) for 5-7 days.
- Observe the patient for 45 minutes post line flush before discharge.

Ongoing Management

- Daily clinical examination, CRP and reassessment.
- In general, all cases should be discussed with Microbiology and you must ensure any positive results are discussed with them

Stopping antibiotics

- Patients should always be discussed with a consultant before stopping

All known oncology patients who are seen or who contact the NDDH Paediatric team with a query should be discussed with both the NDDH Paediatric Consultant On-Call **and** either Dr Hayes or Dr Parke at the RD&E (contactable via Bramble Ward) or if they are not available the Paediatric Oncology team at Bristol Children's Hospital (any new patients presenting with an oncological diagnosis should be discussed with the Bristol team).

Useful numbers are listed below (Appendix 2)

When the patient has been stabilised appropriate transfer (usually to the RD&E) needs to be arranged. These patients are at risk of life threatening sepsis therefore transfer needs to be by ambulance with, at least, a paramedic crew as an escort.

Useful information about our shared Oncology patients including antibiotic and transfusion requirements is in the Minutes of the weekly RD&E MDT "weekly Oncology MDT minutes" e-mailed to all Paediatric Consultants and saved in the Paediatric Drive – Oncology MDT minutes.

Depending on the results of blood tests blood and platelet transfusions may be necessary, please give our blood transfusion laboratory as much notice as possible. Contact the Paediatric Oncology Team if any queries regarding patient specific transfusion requirements.

Formula for calculating the blood volume to transfuse (Appendix 1)

Other relevant Guidelines accessed from Bristol Children's Hospital Oncology Guideline site are listed below:

- Acutely Unwell Haem/Onc patient Management
- Brain and Spinal Cord Tumour Management
- Encephalopathy Management in Haem/Onc patients
- Haematology and Oncology New Patient Checklist
- Leukaemia: management of initial high WCC
- Spinal Cord Compression Management in Haem/Onc patients
- SVC Obstruction Management in Haem/Onc patients
- Tumour Lysis Syndrome

Appendix 1:

The volume of packed red cells to be transfused should be calculated according to the formula below and always prescribed in mls not units:

Weight of the patient (kg) X Rise of Hb (g/L) ÷ 10 X 4 = Volume to be Transfused (mls)

Establish with a senior doctor the aimed for final Hb concentration and current Hb.

e.g. Calculation for rise of Hb needed if current Hb 70g/L and aimed for Hb 120g/L:

120 - 70 = 50g/L is the aimed for rise.

15kg (patient weight) x 50g/L (rise of Hb) x 4 = 300mls to be transfused

Do not give more than 20ml/kg per transfusion

unless actively bleeding.

Red cells should be transfused within 4 hours following removal from cold storage.

Appendix 2: Useful numbers

NDDH:

Jenny Birch (Clic nurse)	0776 7308469
Consultant Paediatrician	Bleep 340. Via Switchboard out of hours

Royal Devon and Exeter Hospital:

Dr Corinne Hayes or Dr Simon Parke	01392 402684 / 402685
or via Bramble ward	01392 402679 / 402678
R,D and E Oncology team e-mail:	rde-tr.PaediatricOncology@nhs.net

Out of hours first point of contact:

On-call Paediatric Registrar via Bramble Yellow Ward, also liaise with the Paediatric Oncology Nurse on shift at Bramble Ward at the RD&E

Bristol Children's Hospital: 0117 9230000

Contact Oncology Day beds or on-call Oncology Registrar / Oncology Consultant on-call